Neurobehavioral Factors in Eating Disorders

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Introduction

This report analyzes neurobehavioral factors associated with **self-perceived eating disorder status** among university students in Bangladesh. The dataset contains survey responses from **550 students** and includes measures related to cognitive restraint, emotional eating, food cue responsiveness, body image concerns, and related perceptions.

Primary outcome: `perception_eating_disorder` (binary; 0 = No, 1 = Yes).

Approach: I created composite domain scores and evaluated associations using group comparisons and logistic regression. A simplified model was selected for interpretability and comparable predictive performance.

Key findings (final model):

- **Affective (Emotional Eating)** was the strongest predictor (large odds ratio; substantial risk gradient across quartiles).
- **Cognitive Control/Restriction** showed a strong independent association with perceived eating disorder status.
- **Body Image Distress** (reverse-coded so **higher = worse** distress) remained a significant predictor.
- Predictive performance was good on a test set (**AUC ~ 0.82**), with balanced classification performance at a 0.5 probability threshold.

Practical implication: The model is suitable as a **screening-oriented decision-support tool** (not diagnostic), highlighting students who may benefit from follow-up assessment or targeted interventions (emotion regulation, healthy eating patterns, body image support).

Phase One: Ask

Eating disorders are shaped by a mix of brain and behavior factors, like how people handle their emotions, how they react to food cues, their efforts to restrict eating, and worries about body image. College students might be especially at risk because of the stress they face, big changes in their daily lives, and the pressure from social and environmental factors.

Business Task:

The goal of this analysis is to identify the **primary neurobehavioral drivers** of eating disorders among university students in Bangladesh. By analyzing **550** survey responses across seven critical factors, I aim to provide university health officials with data-driven insights to support the creation of targeted mental health interventions.

Key Stakeholders:

- **Primary Audience:** University administration, student health services, or mental health researchers.
- **Stakeholder Goal:** To design better prevention programs or early intervention strategies based on the "*brain-based models*" I mentioned in dataset description.

Success Metrics

1. **Dependent Variable (Target):** Self-perception about eating disorders.
2. **Independent Variables (Predictors):** Scores for Cognitive Control, Affective Value, Salience/Taste, Body Image, and Reward Processing.

Phase Two: Prepare

Data Source & Organization

The dataset used in this project contains survey responses related to neurobehavioral factors associated with self-perceived eating disorder status among university students in Bangladesh.

Files used in the project

Two versions of the dataset are maintained:

- **Raw dataset:** `data/raw/neurobehavioral_factors_in_eating_disorders_raw.csv`
- **Cleaned dataset:** `data/clean/neurobehavioral_factors_in_eating_disorders_clean.csv`

The raw file is preserved as the original source, while the cleaned file is used for scoring, modeling, and reporting.

Data structure

The dataset is organized in flat-file tabular format, where:

- each row represents one participant
- each column represents one survey variable or derived feature

The analysis in this project uses **550 student responses**.

Variables included

The dataset includes variables related to:

- demographic characteristics
- cognitive control / restriction behaviors
- emotional eating tendencies
- food cue responsiveness / salience
- habitual eating patterns
- body image concerns
- self-perceived eating disorder status

Data credibility considerations

Because this is survey-based research, the dataset is useful for identifying patterns and associations, but several quality considerations remain important:

Criterion	Status	Assessment
Reliable	Medium	The sample size is adequate for an exploratory university-level analysis, but responses are self-reported and may be affected by recall bias or under-reporting.
Original	To verify	Keep this as "High" only if the dataset documentation confirms it was collected directly from students by the stated institution.
Comprehensive	High	The dataset includes multiple neurobehavioral domains relevant to eating disorder perception, which supports domain-level scoring and multivariable modeling.
Current	To verify	Keep the collection period only if it is explicitly stated in the dataset source documentation.
Cited	Medium to High	The source should be documented clearly in the repository README or data documentation file.

Privacy & Ethics

This project studies a mental health-related outcome, so privacy and ethical handling are important. Before analysis, the dataset should be checked to ensure it does not contain personally identifiable information such as:

- names
- student IDs
- email addresses
- phone numbers

The analysis is intended for research and screening-oriented insight generation only. It should not be interpreted as a clinical diagnostic assessment.

Data readiness for analysis

Before modeling, the dataset was prepared so it could be used consistently in R:

- column names were standardized
- categorical survey responses were converted into numeric scales
- domain-level composite scores were later created from related item groups
- the outcome variable was represented as a binary variable:
  - `0 = No`
  - `1 = Yes`

The cleaned dataset used for analysis is:

`data/clean/neurobehavioral_factors_in_eating_disorders_clean.csv`

Phase Three: Process

In this section I document *how the cleaned, analysis-ready dataset was produced*, the tools used, and the quality checks performed. The goal is reproducibility: a common reader should understand what was changed and why.

Tools Used

- **Excel / Google Sheets**: initial cleaning, inspection, and recoding support.
- **R / RStudio**: composite score construction, analysis, modeling, validation, and reporting.

File management and versioning

- Raw dataset was kept unchanged.
- A separate cleaned file was created for analysis.
- Naming convention used:
  - `neurobehavioral_factors_in_eating_disorders_raw.csv` (original)
  - `neurobehavioral_factors_in_eating_disorders_clean.csv` (analysis-ready)

Analysis uses: `neurobehavioral_factors_in_eating_disorders_clean.csv`

Data cleaning steps performed in Excel

I used Excel because the dataset contains a resonable quantity of data.

Column standardization

- I verified all column headers were correctly labeled and consistent.
- Ensured no duplicate or blank column names.
- Removed trailing spaces and accidental duplicates.

Missing values and integrity checks

- Scanned each column for blank cells.
- Confirmed the dataset had one row per participant and no duplicated responses.
- Verified demographic fields had valid categories.

Recoding categorical survey responses into numeric scales

Survey response options (*Never, Seldom, Sometimes, Often*) were converted into numeric codes to allow statistical analysis.

Example frequency coding (as used in this case study):

- `Never` = 0
- `Seldom` = 1
- `Sometimes` = 2
- `Often` = 3

Recoding "days" response ranges into numeric codes

Items measured as frequency across a time window (*1–12 days, 13–27 days, Every day*) were also converted to numeric codes.
During analysis validation, I identified that these day-based variables were oriented such that **lower numeric values represented more frequent distress**. I corrected this later in the *analysis phase* by reverse-coding the composite body-image score.

Creating composite domain scores (aggregation)

To reduce dimensionality and support interpretable analysis, item-level variables were aggregated into composite domain scores. Aggregation was performed in **RStudio**, producing domain scores such as:

- `score_cognitive_control`
- `score_affective`
- `score_salience`
- `score_habit`
- `score_body_image`

During analysis, the body-image score is reverse-coded to create `score_body_image_rev`, so that higher values consistently indicate higher distress.

Aggregation rule: row-wise mean (not sum)

Outcome variable preparation

The outcome column `Perception_EatingDisorder` was converted into a binary numeric variable:

- `0` = does not perceive having an eating disorder
- `1` = perceives having an eating disorder

Final processed dataset used in R

The final dataset imported into R includes:

- Demographic variables
- Numeric-encoded item responses
- Composite domain scores
- Binary outcome variable

File used for analysis: `neurobehavioral_factors_in_eating_disorders_clean.csv`

Reproducibility note

Because processing was done in Excel, formulas and mapping tables are not automatically tracked in code. To improve reproducibility, this project stores:

- the final analysis-ready dataset (`neurobehavioral_factors_in_eating_disorders_clean.csv`)
- exported result tables (`tables/`)
- saved model objects (`models/`)
- generated figures (`figures/`)

Analyze (Exploration, Statistical Testing, and Modeling)

This section answers the core question:

Which neurobehavioral domains are most strongly associated with self-perceived eating disorder status, and how well can a simple, interpretable model predict ED perception?

I follow an evidence-building sequence:

1) Verify outcome and distributions

2) Describe domain scores

3) Test group differences (bivariate)

4) Fit and simplify a logistic regression model (multivariable)

5) Evaluate performance on unseen data (test set)

6) Choose an operational threshold and build stakeholder-friendly risk summaries

Load libraries

```r
# Libraries for data work + plots + tables
library(tidyverse)
library(janitor)
library(knitr)
library(kableExtra)

# Libraries for modeling and evaluation
library(rsample)
library(broom)
library(pROC)
library(yardstick)

set.seed(123)  # makes results reproducible
```

Import the cleaned dataset

This case study starts analysis from a cleaned dataset (categorical responses already converted into numeric codes). I use a project-relative path so the report can be reproduced across environments.

```r
glimpse(df_clean)
```

```
## Rows: 550
## Columns: 57
## $ age_range                                        <chr> "18-24", "25-34", "1~
## $ gender                                           <chr> "Male", "Female", "M~
## $ division_of_origin                               <chr> "Rajshahi", "Chattog~
## $ current_education_level                          <chr> "Bachelor's degree 3~
## $ employment_status                                <chr> "Student", "Self-emp~
## $ marital_status                                   <chr> "In a relationship",~
## $ eat_less_on_weight_gain                          <dbl> 3, 3, 3, 3, 2, 3, 3,~
## $ eat_less_at_mealtime                             <dbl> 0, 3, 0, 3, 0, 3, 3,~
## $ refuse_food_weight_concern                       <dbl> 2, 1, 3, 0, 1, 3, 0,~
## $ monitor_food                                     <dbl> 2, 3, 1, 2, 3, 2, 1,~
## $ eat_slimming_foods                               <dbl> 0, 1, 2, 2, 0, 0, 2,~
## $ eat_less_after_overeating                        <dbl> 3, 3, 3, 3, 3, 2, 0,~
## $ eat_less_to_prevent_weight_gain                  <dbl> 1, 3, 3, 2, 3, 0, 3,~
## $ avoid_snacks_between_meals_for_weight_concern    <dbl> 3, 0, 2, 3, 3, 0, 1,~
## $ avoid_evening_eating_for_weight_concern          <dbl> 1, 2, 1, 2, 3, 0, 0,~
## $ consider_weight_when_eating                      <dbl> 3, 1, 0, 1, 2, 2, 3,~
## $ eat_when_unoccupied                              <dbl> 2, 2, 3, 0, 0, 3, 1,~
## $ eat_when_irritated                               <dbl> 2, 3, 3, 2, 2, 0, 0,~
## $ eat_when_depressed_or_discouraged                <dbl> 2, 3, 3, 0, 2, 1, 0,~
## $ eat_when_lonely                                  <dbl> 3, 1, 3, 0, 1, 0, 3,~
## $ eat_when_someone_let_you_down                    <dbl> 3, 3, 2, 2, 2, 2, 2,~
## $ eat_when_angry                                   <dbl> 0, 0, 3, 3, 1, 1, 3,~
## $ eat_when_expecting_bad                           <dbl> 1, 1, 3, 3, 0, 3, 0,~
## $ eat_when_anxious                                 <dbl> 1, 3, 2, 2, 1, 2, 2,~
## $ eat_when_things_go_wrong                         <dbl> 1, 3, 0, 0, 1, 1, 3,~
## $ eat_when_frightened                              <dbl> 2, 3, 2, 3, 1, 1, 2,~
```

```
## $ eat_when_disappointed                              <dbl> 1, 2, 3, 3, 2, 0, 2,~
## $ eat_when_emotionally_upset                         <dbl> 0, 3, 1, 1, 0, 0, 3,~
## $ eat_when_bored_or_restless                         <dbl> 0, 2, 0, 1, 1, 0, 3,~
## $ eat_more_if_food_tasty                             <dbl> 3, 3, 3, 3, 2, 3, 3,~
## $ eat_more_if_food_smells_or_looks_good              <dbl> 0, 2, 0, 2, 2, 3, 2,~
## $ eat_when_see_delicious_food                        <dbl> 2, 1, 3, 0, 1, 3, 0,~
## $ eat_delicious_food_immediately                     <dbl> 2, 3, 1, 2, 1, 2, 1,~
## $ desire_to_buy_from_bakery                          <dbl> 0, 1, 2, 2, 0, 0, 2,~
## $ desire_to_buy_from_snack_bar_or_cafe               <dbl> 3, 3, 3, 3, 3, 2, 0,~
## $ desire_to_eat_when_see_others_eating               <dbl> 1, 3, 3, 2, 3, 0, 3,~
## $ resist_delicious_food                              <dbl> 3, 0, 2, 2, 3, 0, 1,~
## $ eat_more_when_see_others_eating                    <dbl> 1, 2, 1, 2, 3, 0, 0,~
## $ eat_when_preparing_meal                            <dbl> 3, 1, 0, 1, 3, 2, 3,~
## $ eat_specific_foods_habitually                      <dbl> 3, 3, 0, 2, 3, 0, 0,~
## $ location_triggers_habitual_eating                  <dbl> 3, 2, 2, 3, 3, 2, 2,~
## $ automatic_eating_when_experiencing_strong_emotion <dbl> 2, 0, 0, 3, 1, 2, 0,~
## $ realize_after_eating_out_of_habit                  <dbl> 0, 0, 0, 3, 3, 1, 1,~
## $ days_desire_for_flat_stomach                       <dbl> 2, 1, 3, 1, 1, 2, 2,~
## $ days_felt_fat                                      <dbl> 3, 1, 1, 1, 2, 2, 1,~
## $ days_weight_affected_self_judgment                 <dbl> 1, 1, 1, 1, 1, 0, 2,~
## $ days_shape_affected_self_judgment                  <dbl> 3, 0, 2, 1, 1, 3, 0,~
## $ days_dissatisfied_with_weight                      <dbl> 1, 2, 1, 1, 1, 2, 1,~
## $ days_dissatisfied_with_shape                       <dbl> 1, 3, 1, 2, 1, 2, 0,~
## $ days_uncomfortable_to_see_own_body                 <dbl> 1, 0, 3, 2, 3, 0, 2,~
## $ days_uncomfortable_others_seeing_your_shape        <dbl> 1, 0, 1, 0, 1, 2, 2,~
## $ days_tried_limit_food_to_control_shape_or_weight   <dbl> 2, 1, 3, 3, 1, 2, 2,~
## $ days_fasted_to_control_shape_or_weight             <dbl> 3, 2, 1, 1, 2, 2, 0,~
## $ days_excluded_food_to_control_shape_or_weight      <dbl> 1, 2, 1, 3, 1, 0, 2,~
## $ days_followed_rules_to_control_shape_or_weight     <dbl> 3, 0, 2, 1, 1, 3, 0,~
## $ days_fear_losing_control_over_eating               <dbl> 1, 2, 1, 1, 1, 2, 3,~
## $ perception_eating_disorder                         <dbl> 1, 1, 1, 1, 1, 0, 1,~
```

I confirm composite domain scores exist (or compute them)

To keep the analysis consistent and reproducible, I use one fixed scoring specification for each neurobehavioral domain. Each score is calculated as the row-wise mean of its component items.

```r
# Explicit item groups used to build each domain score
cognitive_items <- c(
  "eat_less_on_weight_gain",
  "eat_less_at_mealtime",
  "refuse_food_weight_concern",
  "monitor_food",
  "eat_slimming_foods",
  "eat_less_after_overeating",
  "eat_less_to_prevent_weight_gain",
  "avoid_snacks_between_meals_for_weight_concern",
  "avoid_evening_eating_for_weight_concern",
  "consider_weight_when_eating"
)

affective_items <- c(
  "eat_when_unoccupied",
```

```r
  "eat_when_irritated",
  "eat_when_depressed_or_discouraged",
  "eat_when_lonely",
  "eat_when_someone_let_you_down",
  "eat_when_angry",
  "eat_when_expecting_bad",
  "eat_when_anxious",
  "eat_when_things_go_wrong",
  "eat_when_frightened",
  "eat_when_disappointed",
  "eat_when_emotionally_upset",
  "eat_when_bored_or_restless"
)

salience_items <- c(
  "eat_more_if_food_tasty",
  "eat_more_if_food_smells_or_looks_good",
  "eat_when_see_delicious_food",
  "eat_delicious_food_immediately",
  "desire_to_buy_from_bakery",
  "desire_to_buy_from_snack_bar_or_cafe",
  "desire_to_eat_when_see_others_eating",
  "resist_delicious_food",
  "eat_more_when_see_others_eating",
  "eat_when_preparing_meal"
)

habit_items <- c(
  "eat_specific_foods_habitually",
  "location_triggers_habitual_eating",
  "automatic_eating_when_experiencing_strong_emotion",
  "realize_after_eating_out_of_habit"
)

body_image_items <- c(
  "days_desire_for_flat_stomach",
  "days_felt_fat",
  "days_weight_affected_self_judgment",
  "days_shape_affected_self_judgment",
  "days_dissatisfied_with_weight",
  "days_dissatisfied_with_shape",
  "days_uncomfortable_to_see_own_body",
  "days_uncomfortable_others_seeing_your_shape",
  "days_tried_limit_food_to_control_shape_or_weight",
  "days_fasted_to_control_shape_or_weight",
  "days_excluded_food_to_control_shape_or_weight",
  "days_followed_rules_to_control_shape_or_weight",
  "days_fear_losing_control_over_eating"
)

# Check that all required columns exist before scoring
required_items <- c(
  cognitive_items,
```

```r
  affective_items,
  salience_items,
  habit_items,
  body_image_items
)

stopifnot(all(required_items %in% names(df_clean)))

# Build composite scores as row-wise means
df_scored <- df_clean %>%
  mutate(
    score_cognitive_control = rowMeans(select(., all_of(cognitive_items)), na.rm = TRUE),
    score_affective         = rowMeans(select(., all_of(affective_items)), na.rm = TRUE),
    score_salience          = rowMeans(select(., all_of(salience_items)), na.rm = TRUE),
    score_habit             = rowMeans(select(., all_of(habit_items)), na.rm = TRUE),
    score_body_image        = rowMeans(select(., all_of(body_image_items)), na.rm = TRUE)
  )
```

Reverse the body image score (higher = worse)

The body-image items are coded so that lower values reflect more frequent distress. I reverse the composite score so that higher values consistently indicate higher concern or higher risk across domains.

```r
days_max <- max(unlist(df_clean[body_image_items]), na.rm = TRUE)

df_scored <- df_scored %>%
  mutate(score_body_image_rev = days_max - score_body_image)
```

I do a quick check:

```r
df_scored %>%
  select(
    score_cognitive_control,
    score_affective,
    score_salience,
    score_habit,
    score_body_image,
    score_body_image_rev
  ) %>%
  summary()
```

```
##  score_cognitive_control score_affective  score_salience  score_habit
##  Min.   :0.600           Min.   :0.6154   Min.   :0.60   Min.   :0.000
##  1st Qu.:1.400           1st Qu.:1.4615   1st Qu.:1.30   1st Qu.:1.250
##  Median :1.600           Median :1.6923   Median :1.60   Median :1.500
##  Mean   :1.633           Mean   :1.6534   Mean   :1.62   Mean   :1.551
##  3rd Qu.:1.900           3rd Qu.:1.8462   3rd Qu.:1.90   3rd Qu.:2.000
##  Max.   :2.700           Max.   :2.6154   Max.   :2.60   Max.   :3.000
##  score_body_image score_body_image_rev
##  Min.   :0.3846   Min.   :0.3077
##  1st Qu.:1.1538   1st Qu.:1.3077
##  Median :1.3846   Median :1.6154
```

```
##  Mean   :1.4434   Mean   :1.5566
##  3rd Qu.:1.6923   3rd Qu.:1.8462
##  Max.   :2.6923   Max.   :2.6154
```

Define the final analysis dataset

I keep the outcome variable numeric for modeling and create a separate labeled version for plots and group comparisons.

```r
score_vars <- c(
  "score_cognitive_control",
  "score_affective",
  "score_salience",
  "score_habit",
  "score_body_image_rev"
)

stopifnot(all(score_vars %in% names(df_scored)))

df_analysis <- df_scored %>%
  mutate(
    perception_eating_disorder = as.integer(perception_eating_disorder)
  )

stopifnot(all(na.omit(df_analysis$perception_eating_disorder) %in% c(0, 1)))

df_grouped <- df_analysis %>%
  mutate(
    ed_group = factor(
      perception_eating_disorder,
      levels = c(0, 1),
      labels = c("No", "Yes")
    )
  )
```

Validate the outcome variable

Before analysis, I confirm that the outcome is binary and check how many students fall into each group.

- `0` = does not perceive an eating disorder
- `1` = perceives an eating disorder

Table 2: Outcome distribution: Perception of Eating Disorder (0 = No, 1 = Yes)

ED perception	Count (n)	Proportion
0	254	0.462
1	296	0.538

Summary statistics for the scores

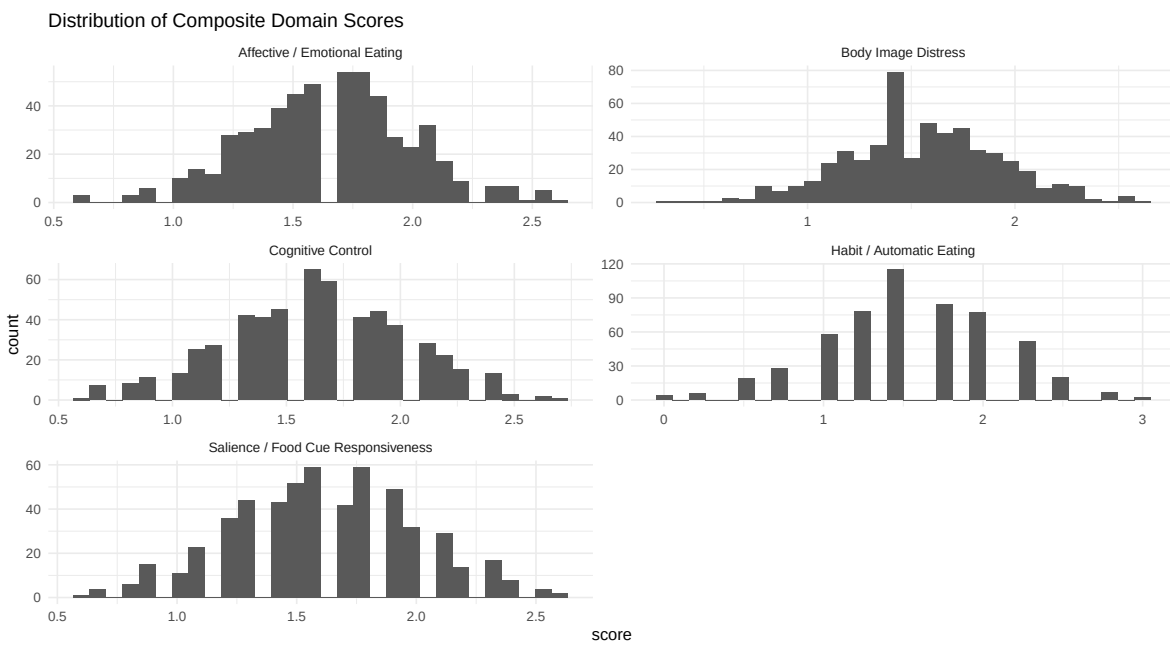
This gives a simple overview of average levels and variation.

```
df_analysis %>%
  select(all_of(score_vars)) %>%
  summary()
```

```
##  score_cognitive_control score_affective  score_salience  score_habit
##  Min.   :0.600           Min.   :0.6154   Min.   :0.60   Min.   :0.000
##  1st Qu.:1.400           1st Qu.:1.4615   1st Qu.:1.30   1st Qu.:1.250
##  Median :1.600           Median :1.6923   Median :1.60   Median :1.500
##  Mean   :1.633           Mean   :1.6534   Mean   :1.62   Mean   :1.551
##  3rd Qu.:1.900           3rd Qu.:1.8462   3rd Qu.:1.90   3rd Qu.:2.000
##  Max.   :2.700           Max.   :2.6154   Max.   :2.60   Max.   :3.000
##  score_body_image_rev
##  Min.   :0.3077
##  1st Qu.:1.3077
##  Median :1.6154
##  Mean   :1.5566
##  3rd Qu.:1.8462
##  Max.   :2.6154
```

Plot the score distributions

These plots show whether the final domain scores are reasonably spread out.



Compare ED=0 vs ED=1 (simple group test)

I compare average domain scores between students who do and do not perceive having an eating disorder.

Table 3: Group differences (t-tests): ED=0 vs ED=1

domain	mean_no	mean_yes	p_value
Cognitive Control	1.468898	1.773986	0.0000000
Affective / Emotional Eating	1.514537	1.772609	0.0000000
Salience / Food Cue Responsiveness	1.485433	1.735135	0.0000000
Habit / Automatic Eating	1.520669	1.577703	0.2203215
Body Image Distress	1.465778	1.634615	0.0000005

I build the final logistic regression model

A logistic regression predicts a binary outcome (0/1). I use the three strongest domains we found earlier:

- Emotional eating (`score_affective`)
- Cognitive control (`score_cognitive_control`)
- Body image distress (`score_body_image_rev`)

Train/test split

I split the data into a training set (70%) and a test set (30%), while preserving the outcome balance across groups.

```
set.seed(123)

split_obj <- initial_split(
  df_analysis,
  prop = 0.7,
  strata = perception_eating_disorder
)

train <- training(split_obj)
test  <- testing(split_obj)
```

Fit model (train set)

A logistic regression is appropriate because the outcome is binary (`0/1`). I use the three strongest and most interpretable domains from the earlier analysis.

```
m_simple <- glm(
  perception_eating_disorder ~
    score_affective +
    score_cognitive_control +
    score_body_image_rev,
  data = train,
  family = binomial()
)

summary(m_simple)
```

```
##
## Call:
## glm(formula = perception_eating_disorder ~ score_affective +
```

```
##     score_cognitive_control + score_body_image_rev, family = binomial(),
##     data = train)
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -9.8092     1.1288  -8.690  < 2e-16 ***
## score_affective           2.6377     0.4150   6.355 2.08e-10 ***
## score_cognitive_control   2.2754     0.3518   6.468 9.96e-11 ***
## score_body_image_rev      1.2327     0.3228   3.819 0.000134 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 529.99  on 383  degrees of freedom
## Residual deviance: 404.06  on 380  degrees of freedom
## AIC: 412.06
##
## Number of Fisher Scoring iterations: 4
```

Odds ratios table (easier interpretation)

Odds ratios make the model easier to interpret: values above 1 indicate higher odds of ED perception as the predictor increases.

Table 4: Final model odds ratios (higher OR = higher risk)

term	estimate	conf.low	conf.high	p.value
Affective (Emotional Eating)	13.980811	6.376340	32.565786	0.000000
Cognitive Control/Restriction	9.732103	4.990067	19.892653	0.000000
Body Image Distress	3.430477	1.846086	6.564169	0.000134

Evaluate the model (test set)

Predict probabilities

```r
test <- test %>%
  mutate(
    pred_prob = predict(m_simple, newdata = test, type = "response")
  )
```

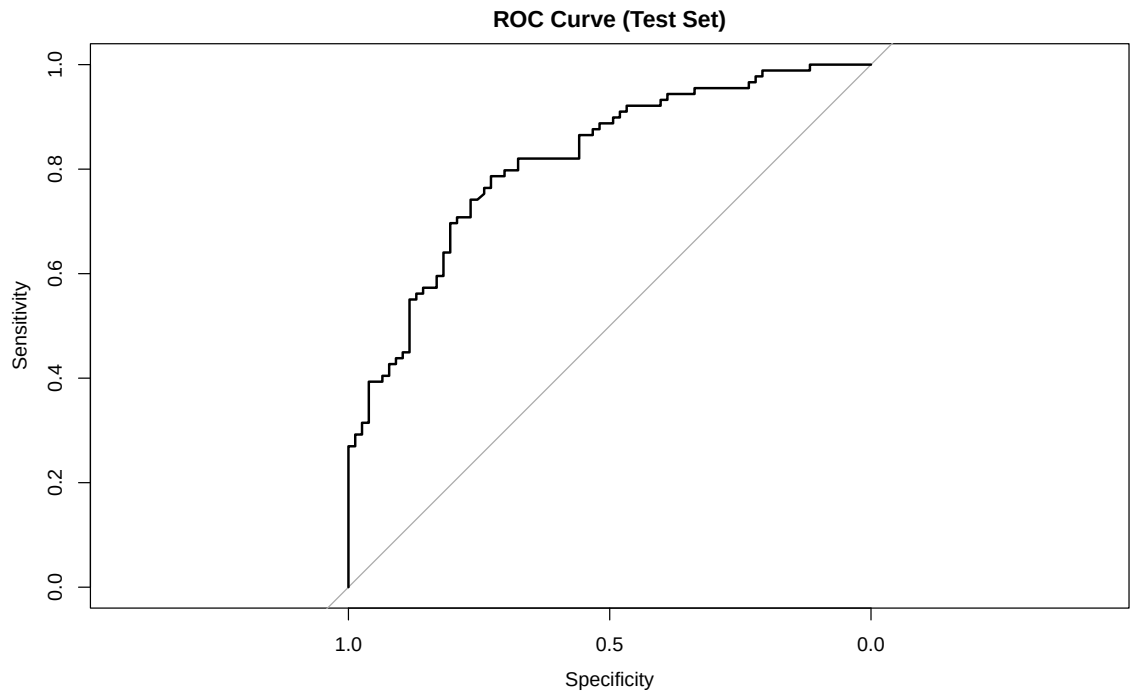
ROC curve and AUC

The ROC curve shows how well the model separates ED=1 from ED=0 across all possible thresholds. The AUC summarizes that discrimination into one number.

```r
roc_obj <- pROC::roc(
  response = test$perception_eating_disorder,
  predictor = test$pred_prob
)

auc_value <- as.numeric(pROC::auc(roc_obj))
auc_value
```

```
## [1] 0.8156282
```



Confusion matrix (threshold = 0.50)

To create a practical classification rule, I classify cases with predicted probability greater than or equal to 0.50 as ED=1.

```r
final_t <- 0.50

truth <- factor(test$perception_eating_disorder, levels = c(0, 1))
pred  <- factor(if_else(test$pred_prob >= final_t, 1L, 0L), levels = c(0, 1))

table(Truth = truth, Predicted = pred)
```

```
##      Predicted
## Truth  0  1
##     0 56 21
##     1 21 68
```

Key metrics (event = ED=1)

These metrics summarize how the model performs at the chosen 0.50 threshold.

Table 5: Model performance on test set (threshold = 0.50)

threshold	accuracy	sensitivity	specificity	precision
0.5	0.747	0.764	0.727	0.764

Risk summary (quartiles)

To make the results easier to explain, I compare predicted risk between lower-score and higher-score groups for each key domain.

Table 6: Predicted risk: low (Q1) vs high (Q4) for each key domain

aff_q1	aff_q4	cog_q1	cog_q4	body_q1	body_q4
0.2840481	0.763228	0.2604202	0.7653002	0.3632788	0.6630594

Final takeaways

Key findings from the final model

- **Emotional eating** (`score_affective`) is the strongest predictor: students with higher emotional-eating scores are much more likely to report that they believe they have an eating disorder.
- **Cognitive control / restriction** (`score_cognitive_control`) is also strongly related to ED perception: more dieting/restriction behaviors are linked to higher likelihood of ED perception.
- **Body image distress** (`score_body_image_rev`) adds independent value: students with higher body image distress are more likely to report ED perception.

Model quality (how well it predicts)

- The **AUC** summarizes how well the model separates ED=1 from ED=0 across all classification thresholds.
- At a **0.50 threshold**, the model provides a practical balance between:
  - **Sensitivity** (catching ED=1 cases)
  - **Specificity** (correctly ruling out ED=0 cases)

Important limitation (interpretation rule)

This is a **cross-sectional survey**:

- The results show **association**, not causation.
- The model is best used as a **screening support tool**, not a diagnostic instrument.

Share (Visualizations and Key Findings)

In this phase I communicate the results clearly to different stakeholders (university health services, administration, and researchers).
The goal is to answer: **What did I find, why does it matter, and what should be done next?**

Key findings (headline statements)

- Students with higher **emotional eating** scores are substantially more likely to report ED perception.
- **Cognitive restriction/control** and **body image distress** are also independently associated with ED perception.
- The final logistic regression model shows **good discrimination** on the test set (AUC reported below).

- At a 0.50 cutoff, the model balances sensitivity and specificity reasonably well for general reporting.

This section presents five stakeholder-ready outputs derived from the analysis above:

- Outcome distribution (context)
- Group differences plot (ED=0 vs ED=1)
- Final model odds ratios (interpretation)
- Model performance (ROC/AUC + confusion matrix metrics)
- Risk gradient plot (what changes from low to high scores)

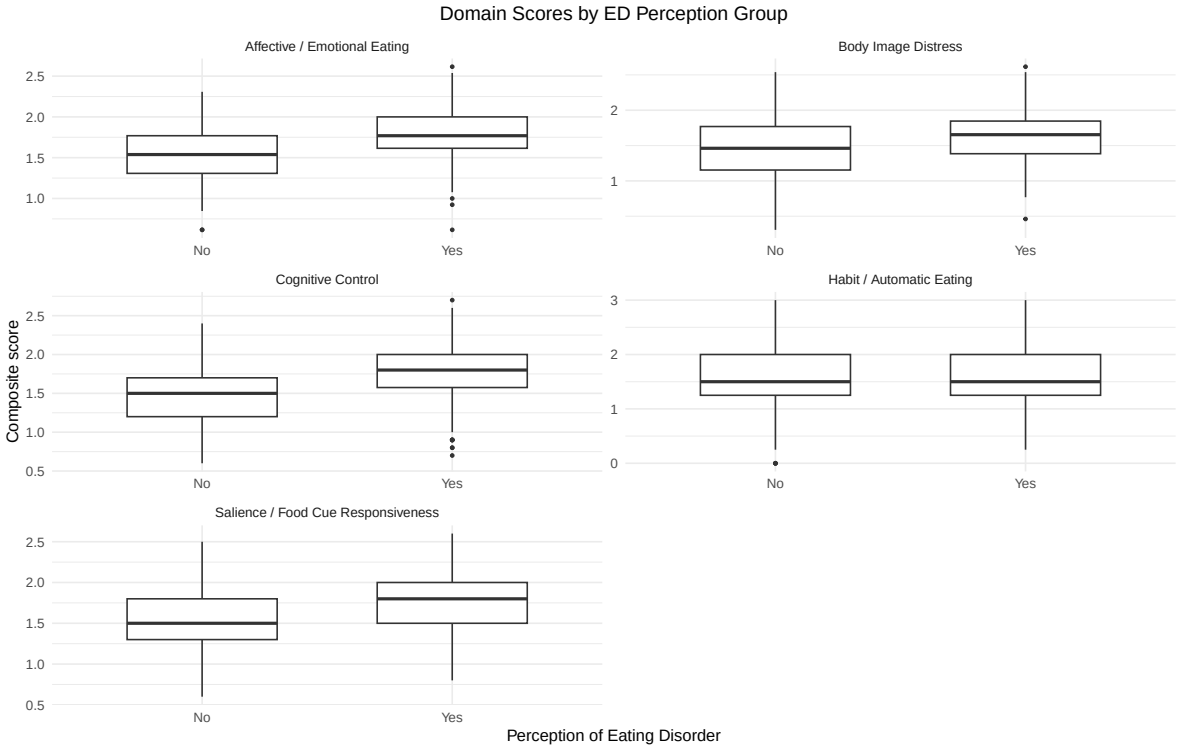
Visual 1 - Outcome distribution (context)

Table 7: Outcome distribution (0 = No, 1 = Yes)

ED perception	Count (n)	Percent (%)
0	254	46.2
1	296	53.8

Visual 2 — Group differences in domain scores (ED=0 vs ED=1)

This figure shows how the final domain scores differ between students who do and do not perceive having an eating disorder.



Visual 3 — Final model results (Odds Ratios)

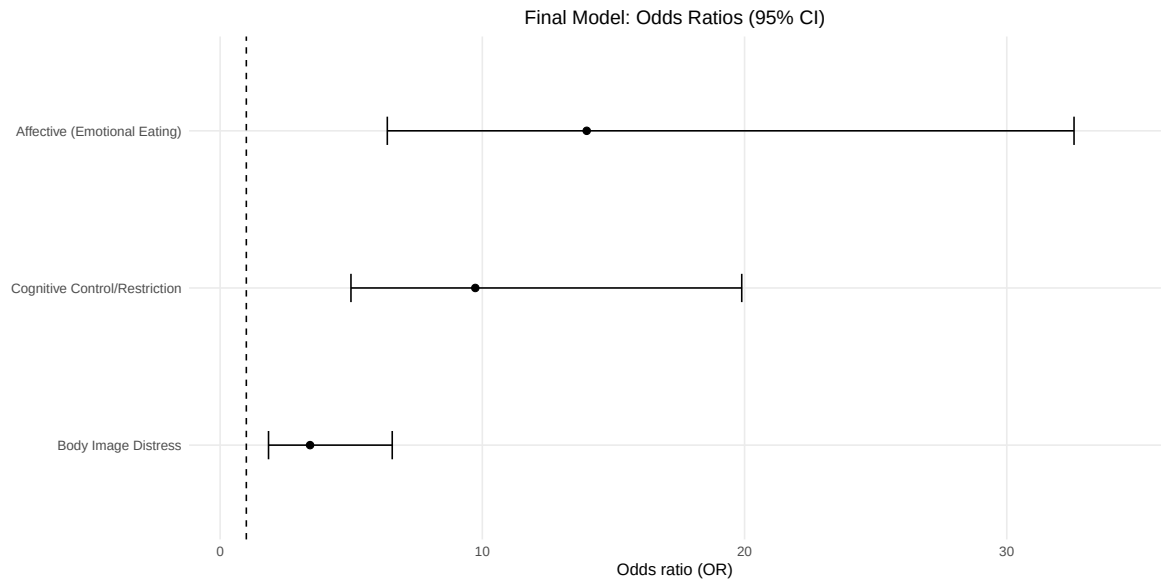
Odds ratios show how strongly each domain is associated with higher odds of ED perception in the final model.

Odds ratio table

Table 8: Final model odds ratios (95% CI)

Predictor	OR	CI low	CI high	p-value
Affective (Emotional Eating)	13.98	6.38	32.57	<0.001
Cognitive Control/Restriction	9.73	4.99	19.89	<0.001
Body Image Distress	3.43	1.85	6.56	<0.001

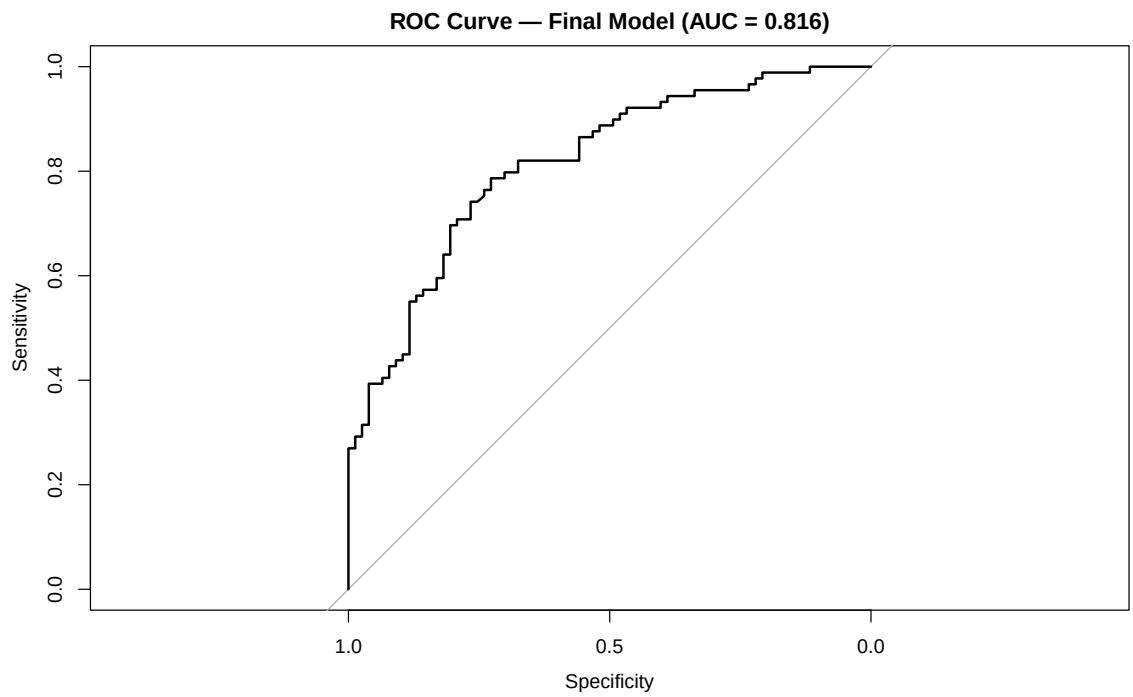
Odds ratio plot



Visual 4 — Model performance (ROC/AUC + metrics)

ROC curve and AUC

The ROC curve shows threshold-free discrimination, while the metrics table summarizes performance at the chosen 0.50 threshold.



Confusion matrix (threshold = 0.50)

```
table(Truth = truth, Predicted = pred)
```

```
##      Predicted
## Truth  0  1
##     0 56 21
##     1 21 68
```

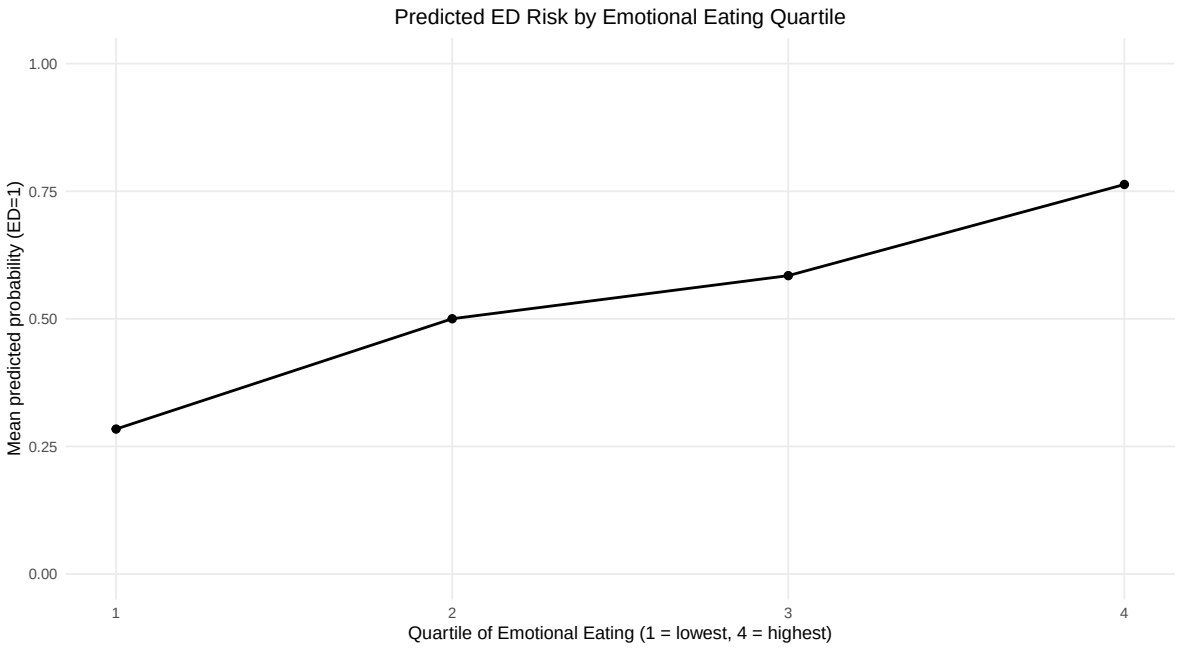
Core metrics (threshold = 0.50)

Table 9: Test set performance (threshold = 0.50, event = ED=1)

threshold	accuracy	sensitivity	specificity	precision
0.5	0.747	0.764	0.727	0.764

Visual 5 — Risk gradient (stakeholder interpretability)

This plot shows how predicted ED risk changes from lower to higher emotional-eating quartiles.



Act (Conclusions, Recommendations, and Next Steps)

- **What is the final conclusion?**
- **How can stakeholders use these insights?**
- **What should be done next, and why?**

Final conclusion (what the analysis supports)

Based on the domain-level modeling and validation:

- **Emotional eating (Affective domain)** is the strongest predictor of self-perceived eating disorder status.
- **Cognitive control/restriction** is also strongly associated with ED perception.
- **Body image distress** adds meaningful independent information.
- The final model shows **good discrimination** on the test set (AUC reported in Phase Five), indicating it can separate higher-risk vs lower-risk students better than chance.

Interpretation rule: These are **associations** from a cross-sectional survey. They do not prove that one factor causes another.

What these findings can support

- **Early screening** (identify students who may benefit from follow-up support)
- **Program design** (target the most influential domains)
- **Resource allocation** (prioritize interventions with the highest expected impact)

This model should be treated as **decision support**, not a clinical diagnostic tool.

Recommended actions (specific and practical)

Recommendation 1 — Target emotional eating (highest-impact domain)

- **What to do:** Implement workshops or counseling modules focused on emotion regulation and stress coping skills (e.g., coping strategies for anxiety, low mood, anger, disappointment).
- **Why:** Emotional eating had the strongest association and steepest risk gradient in the model.
- **Who should implement:** University counseling services in partnership with student affairs.
- **When:** Pilot within one academic term; evaluate uptake and engagement.

Recommendation 2 — Address restrictive eating behaviors and dieting cycles

- **What to do:** Provide psychoeducation about restrictive dieting patterns and encourage balanced eating strategies (aligned with nutrition guidance).
- **Why:** Cognitive control/restriction remained an independent predictor in the final model.
- **Who should implement:** Health services and nutrition educators.
- **When:** Integrate into student wellness programs, orientation sessions, and counseling referrals.

Recommendation 3 — Create body image support programming

- **What to do:** Offer body image resilience sessions and support groups, focusing on self-judgment and dissatisfaction frequency.
- **Why:** Body image distress added predictive value even after including other domains.
- **Who should implement:** Counseling services and trained facilitators.
- **When:** Offer recurring sessions; prioritize high-stress periods (exams).

Risks and limitations (what could go wrong)

- **False positives:** some students flagged by the model may not have an eating disorder.
- **False negatives:** some students may not be flagged even if at risk.
- **Self-report bias:** students may under-report sensitive behaviors.
- **Generalizability:** data is from one university, so performance may change elsewhere.

Next steps (recommended improvements)

External validation

Run the same model on a second student sample (different university or year cohort).
Goal: confirm the model generalizes.

Longitudinal data collection

Collect follow-up responses over time.
Goal: test whether these domains predict future outcomes (stronger evidence than cross-sectional).

Improve measurement reliability

If any domain had low internal consistency, refine item selection or apply psychometric validation.
Goal: ensure composites measure the intended constructs.

Build a simple screening rubric

Translate predicted probabilities into "Low / Medium / High" risk bands.
Goal: make the results easier to apply in practice.

Act deliverables (what is produced at the end)

- A validated, interpretable logistic regression model (`m_simple`)
- A stakeholder-friendly summary of key predictors and performance
- A recommended threshold policy for balanced vs screening use
- A set of actionable intervention recommendations tied directly to model findings